**GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA**
**(Academic Section)**

AS/81/2675 Dated: 23/06/2026

**Circular**

**All HoDs (UG-Engineering)**

Students of UG-Engineering are required to fill and lock choices in the order of preference ('1' being the best choice) of Open Elective-II course for 7th Semester for Academic session **July-Dec, 2026** as mentioned below. Option "Fill Open Elective-II Choice" is available under "**Examination Panel**" Tab at ***https://academics.gndec.ac.in***.

The details of **Open Elective –II (for 7th Semester)** Courses are as under:

| Sr. No. | Course Code | Paper ID | Course Title | Applicable UG Programs | Offering Deptt. | Remarks |
|---|---|---|---|---|---|---|
| 1. | OECE-105 | 17347 | Environmental Impact Assessment | CSE, EE, ECE, IT, ME | CE | -- |
| 2. | OEIT-104 | 17885 | Artificial Intelligence | CE, ECE, ME | IT | -- |
| 3. | OECS-114 | 17614 | Cloud Computing-II | CE, EE, ECE, ME | CSE | To be offered through AWS (Amazon Web Services) Academy. |
| 4. | OEEC-111 | 17783 | e- Waste Recycling and Management | CE, EE, ME, CSE,IT | ECE | -- |
| 5. | OEME-112 | 17693 | Entrepreneurship Practices | CE, CSE, EE,ECE, IT | M&PE | In collaboration with Wadhwani Foundation under National Entrepreneurship Network (NEN) Initiative. ** |
| 6. | OEEE-106 | 17782 | Energy Efficient Machines | CE, ECE, ME, CSE, IT | EE | -- |

**This course will enable a student to build final pitch deck by the end of course. The students will develop business sense as they build on their idea to MVP to Business Planning.

***Syllabi of above mentioned open electives are available on College Notice Board.***

It is mandatory for the students to fill three choices for Open Elective-II priority-wise and lock it by **02.07.2026** up to **3:00PM.** In case a student does not perform this activity in time, then Academic Section will do the allotment without considering student choices.

**Portal for Open Elective choice filling will be operational from 24.06.2026 (10:00AM) to 02.07.2026 (till 3:00PM).**

**Note:**

1. Cloud Computing-II course will be offered only to those students who have studied Cloud Computing –I in 6th Semester as Open Elective-I.
2. After submitting the filled choices, students will not be able to change it.
3. Open Elective-II will be offered only if strength of students is more than or equal to 30 for that particular course.

23/6/26
Assistant Registrar (Academics)

CC:

1. Principal- for information
2. Dean Academics- for information
3. All HoDs (UG Engineering) - to inform concerned students so that choice filling can be done on time.
4. Office copy

**Subject Code:** OECE-105
**Subject Name:** Environmental Impact Assessment

| | |
|---|---|
| **Programme:** B.Tech. (Civil Engineering) | **L:** 3 **T: 0 P:** 0 |
| **Semester: 7/8** | **Teaching Hours:** 36 Hours |
| **Theory/Practical:** Theory | **Credits:** 3 |
| **Internal Marks:** 40 | **Percentage of Numerical/Design Problems:** 20% |
| **External Marks:** 60 | **Duration of End Semester Exam (ESE):** 3 Hours |
| **Total Marks:** 100 | **Course Category:** Open Elective |

**Additional Material Allowed in ESE:** NIL

**On Completion of the course, the student will have the ability to:**

| CO# | Course Outcomes |
|---|---|
| 1. | Identify different aspects related to EIA |
| 2. | Understand the implications related to large scale development projects. |
| 3. | Apply the knowledge gained through the subject to assess the feasibility of a project |
| 4. | Analyze the profitability of a project in terms of environment. |
| 5. | Identify the suitable measures and will be able to prepare a plan for rapid EIA. |
| 6. | Create a plan for environmental impact of a development project. |

**Detailed Contents:**

## Part-A

**Introduction to EIA:** Historical development of Environmental Impact Assessment (EIA). EIA in Project Cycle. Legal and Regulatory aspects in India. – Types and limitations of EIA – Cross sectorial issues and terms of reference in EIA – Public Participation in EIA. EIA Process screening –scoping- setting – analysis – mitigation (6 Hrs)

**Components and methods for EIA:** Matrices – Networks – Checklists – Connections and combinations of processes – Cost benefit analysis – Analysis of alternatives. (8 Hrs)

## Part-B

**Socio-economic impact Assessment:** Definition, Social impact Assessment model and the planning process. Rationale and measurement for SIA variables, Relationship between social impacts and change in Community and Institutional arrangements. Individual and family level impacts. Communities in transition - neighbourhood and Community impacts. Selecting, testing and understanding significant social impacts. Mitigation and enhancement in social assessment. Environmental costing of Projects. (12 Hrs)

**Environmental Management Plan** - preparation, implementation, and review – Mitigation and Rehabilitation Plans – Policy and guidelines for planning and monitoring programs – Post Project audit – Ethical and Quality aspects of Environmental Impact Assessment (6 Hrs)

**Environmental Protection Act:** The EPA, The Water Act, The Air Act, Case Studies, and preparation of environmental impact assessment statement for various industries. (4 Hrs)

**TEXT BOOKS:**


1. Canter, L.W., Environmental Impact Assessment, McGraw Hill, New York.1996
2. Environmental Impact Assessment, Barthwal, R. R. New Age International Publications.


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**REFERENCE BOOKS:**

1. Lawrence, D.P., Environmental Impact Assessment – Practical solutions to current problems, Wiley-Interscience, New Jersey,2003.
2. World Bank –Source book on EIA
3. Petts, J., Handbook of Environmental Impact Assessment, Vol., I and II, Blackwell Science, London,1999.

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Guru Nanak Dev Engineering College, Ludhiana
Department of Information Technology
B. Tech (Information Technology) Scheme 2018

Subject Code: OEIT-104
Subject Name: Artificial Intelligence

| | |
|---|---|
| Programme: B.Tech. | L: 3 T: 0 P: 0 |
| Semester: 7 | Teaching Hours: 36 Hours |
| Theory/Practical: Theory | Credits: 3 |
| Internal Marks: 40 | Percentage of Numerical/Design Problems: 30% |
| External Marks: 60 | Duration of End Semester Exam(ESE): 3hours |
| Total Marks: 100 | Elective Status: Open Elective |

On Completion of the course, the student will have the ability to:

| CO# | Course Outcomes |
|---|---|
| 1. | Understand the concept of Artificial Intelligence, Agents, their types and structure. |
| 2. | Apply and analyze search strategies to solve the informed and uninformed problems |
| 3. | Design and evaluate intelligent expert models for perception and prediction from intelligent environment. |
| 4. | Formulate valid solutions for problems involving uncertain inputs or outcomes by using decision making techniques. |
| 5. | Demonstrate and enrich knowledge to select and apply AI tools to synthesize information and develop models within constraints of application area. |
| 6. | Examine the issues involved in knowledge bases, reasoning systems and planning |

Prerequisites: Knowledge of problem solving using different algorithms and basic programming
Additional Material Allowed in ESE: NIL

Detailed Contents:

Part -A

Introduction:
Intelligence, Foundations of artificial intelligence (AI). History of AI, Agents and Environments, Rationality of Agents, Nature and Structure of Agents, Communication among Agents. [3 L]

Problem Formulation and solution:
State space search, control strategies, heuristic search, problem characteristics, production system characteristics., Generate and test, Hill climbing, best first search, A* search, AO* search, Constraint satisfaction problem, Agenda Driven Search, Mean-end analysis, MinMax Search, Alpha-Beta Pruning, Iterative Deepening. [7 L]

Game playing:
Perfect Information game, Imperfect Information game, Evaluation function, Minimax algorithm, Alpha-beta pruning. [3 L]

Logical Reasoning:
Inference in Propositional logic and First order Predicate logic, Resolution, Logical reasoning, Forward chaining, Backward chaining [3 L]

PART-B

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Knowledge representation:
Mapping between facts and representations, Approaches to knowledge representation, procedural vs declarative knowledge, Forward vs. Backward reasoning, Matching, conflict resolution, Weak and Strong filler structures, semantic nets, frame, conceptual dependency, scripts. [7L]
Planning and Learning Agents:
Intelligent Agents, Nature and structure of Agents, Learning Agents, Introduction to different Forms of Learning, The Planning problem, planning with state space search, partial order planning, planning graphs, planning with propositional logic, Analysis of planning approaches, Hierarchical planning, conditional planning, Continuous and Multi Agent planning.[9L]
Inductive learning:
Decision trees, Rule based learning, Current-best-hypothesis search, Least commitment search, Neural networks, Reinforcement learning, Genetic algorithms. [5L]

Text Books:

1. Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Prentice Hall. 2010
2. Saroj Kaushik, "Artificial Intelligence", Cengage Learning India,2011

Reference Books:

1. Elaine Rich and Kevin Knight, "Artificial Intelligence", Tata McGraw Hill. 2010
2. Trivedi, M.C., "A Classical Approach to Artifical Intelligence", Khanna Publishing House, Delhi.2018
3. David Poole and Alan Mackworth, "Artificial Intelligence: Foundations for Computational Agents", Cambridge University,2010

E-Books and Online Learning Material

1. HandBook of Artificial Intelligence Edited by Avron Barr and Edward A. Feigenbaum, Computer Science Department Stanford University. https://stacks.stanford.edu/file/druid:qn160ck3308/qn160ck3308.pdf

Online Courses and Video Lectures

1. https://www.coursera.org/courses?query=artificial%20intelligence (Accessed on May 20,2020)
2. https://nptel.ac.in/courses/106/105/106105077/ (Accessed on May 20,2020)
3. https://nptel.ac.in/courses/106/102/106102220/ (Accessed on May 20, 2020)
4. https://www.youtube.com/watch?v=bV4t4r3SGuI (Accessed on May 20, 2020)
5. https://www.youtube.com/watch?v=iF1tOCEXLXY (Accessed on May 20,2020)

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**Subject Code :** OECS-114

**Subject Name:** Cloud Computing-II

| | |
|---|---|
| **Programme:** B.Tech (CSE) | **L:** 3 **T:** 0 **P:** 0 |
| **Semester:** 7th | **Teaching Hours:** 36 |
| **Theory / Practical:** Theory | **Credits:** 3 |
| **Internal Marks:** 40 | **Percentage of Numerical/Design/Programming Problems:** 10% |
| **External Marks:** 60 | **Duration of End Semester Exam (ESE) :** 3 hrs |
| **Total Marks:** 100 | **Elective Status :** Open Elective |

**Additional Material Allowed in ESE: [NIL]**

**On Completion of the course, the student will have the ability to:**

| CO# | Course Outcomes (CO) |
|---|---|
| 1 | Apply shared security model on Amazon Web Services (AWS) cloud implementation. |
| 2 | Apply automatic scaling in AWS cloud environments. |
| 3 | Host a static or dynamic website and use Lambda service on AWS cloud. |
| 4 | Use Artificial intelligence and Machine learning services on AWS cloud. |
| 5 | Make use of Internet of Things and Big Data products on AWS cloud. |
| 6 | Implement Blockchain technology using AWS cloud. |

**Detailed Contents:**

**Part-A**

**Amazon Web Services (AWS) Shared Security Model:** Introduction to AWS security model for cloud services, Identity and access management (IAM), Principle of least privilege (PoLP), Denial of service (DoS), Distributed denial of service (DDoS), Watering hole attack, Multi-factor authentication (MFA), Amazon inspector, AWS trusted advisor, Amazon simple storage service (Amazon S3), Amazon elastic block store (Amazon EBS), Amazon relational database service (Amazon RDS)

[06]

**Cloud Services, Instance States and Auto Scaling in Cloud Environments:** Cloud service models: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a

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Service (SaaS). Amazon elastic compute cloud (Amazon EC2), EC2 instance states, AWS instance lifecycle, Instance store volumes, Amazon machine image (AMI), IPv4 address and IPv6 address, Elastic IP address, Automatic scaling in cloud environments, Auto scaling groups, Fleet, Launch template, Scale-out and Scale-in. **[7 hours]**

**Dynamic Web Servers, Lambda and CloudFormation:** Static website, Dynamic website, Amazon CloudFront, Content delivery network (CDN), Edge location, Origin, Distribution, Time to live (TTL), AWS Lambda, CloudFormation template, Infrastructure as code (IaC) **[5 hours]**

**Part-B**

**Artificial Intelligence (AI) and Machine Learning (ML):** Introduction to AI and ML, AWS DeepLens, AI services from AWS platform: Amazon Comprehend, Amazon Forecast, Amazon Lex, Amazon Personalize, Amazon Polly, Amazon Rekognition, Amazon Textract, Amazon Translate, Amazon Transcribe. Impact of AI, Deep learning, Reinforcement learning, Supervised learning, Unsupervised learning, Forecasting, Neural network, AWS machine learning applications. **[8 hours]**

**Internet of Things (IoT) and Big Data:** Introduction to IoT and Big data, AWS IoT services, Apache Hadoop, Big data processing cycle, Data analytics, AWS Big data applications and services. **[6 hours]**

**Blockchain and Cryptocurrency:** Introduction to blockchain technology, Cryptocurrency, Cryptocurrency mining, Decentralized database, Hash, Immutable transactions, Smart contract, AWS blockchain products. **[4 hours]**

**Text Books:**

1. Raj Kumar Buyya, James Broberg, Andrezei Goscinski, "Cloud Computing: Principles and Paradigms", First edition, John Wiley & Sons Inc.
2. Barrie Sosinsky, "Cloud Computing Bible", First edition, Wiley India Pvt. Ltd.

**Reference Books:**

1. John Rittinghouse, James F.Ransome, "Cloud Computing: Implementation, Management, and Security", International edition, CRC Press Taylor and Francis Group.
2. Judith Hurwitz, Robin Bloor, Marcia Kaufman, Fern Halper," Cloud Computing for Dummies", International edition, John Wiley & Sons Inc.

**Books and online learning Material:**

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1 Best Practices for Security, Identity, & Compliance: https://aws.amazon.com/architecture/security-identity-compliance/ Accessed on July 21st, 2022.

2 Share an Amazon EBS Snapshot: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-modifying-snapshot-permissions.html Accessed on July 21st, 2022.

3 Sharing a DB Snapshot: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ShareSnapshot.html Accessed on July 21st, 2022.

4 Bucket Policies and User Policies: https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-iam-policies.html Accessed on July 21st, 2022.

5 AWS Trusted Advisor for Everyone: https://aws.amazon.com/blogs/aws/trusted-advisor-console-basic/ Accessed on July 21st, 2022.

6 Understand the Instance Store and EBS: https://aws.amazon.com/premiumsupport/knowledge-center/instance-store-vs-ebs/ Accessed on July 21st, 2022.

7 Learn About Elastic IP Addresses: https://aws.amazon.com/premiumsupport/knowledge-center/intro-elastic-ip-addresses/ Accessed on July 21st, 2022.

8 https://aws.amazon.com/ec2/autoscaling/ Accessed on July 21st, 2022.

9 AWS Lambda FAQs https://aws.amazon.com/lambda/faqs/ Accessed on July 21st, 2022.

10 Explore AWS AI services https://aws.amazon.com/machine-learning/ai-services/ Accessed on July 21st, 2022.

11 AI with AWS Machine Learning https://aws.amazon.com/ai/ Accessed on July 21st, 2022.

12 https://aws.amazon.com/iot/ Accessed on July 21st, 2022.

13 https://aws.amazon.com/blogs/big-data/ Accessed on July 21st, 2022.

14 Blockchain on AWS https://aws.amazon.com/blockchain/ Accessed on July 21st, 2022.

**Online Courses and Video Lectures**

1 https://youtu.be/bSRTAMPqS3E Accessed on July 21st, 2022.

2 https://www.youtube.com/watch?v=PideBMIcwBQ Accessed on July 21st, 2022.

3 AWS Quick Start - Hosting a Static Website on AWS (Demo) video https://www.youtube.com/watch?v=BpFKnPae1oY Accessed on July 21st, 2022.

4 Getting Started with AWS - Build a Simple Static Website video https://www.youtube.com/watch?v=jHH-rN3y3Qc Accessed on July 21st, 2022.

5 Big Data on Amazon Web Services https://www.youtube.com/watch?v=3fkGCM0feC0&feature=youtu.be Accessed on July 21st, 2022.

6 Introduction to Blockchain on AWS" https://www.youtube.com/watch?v=oMzyZ49pykU accessed on July 21st, 2022.

7 What is Blockchain on AWS?" https://www.youtube.com/watch?v=9xbtq362Scs Accessed on July 21st, 2022.

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# Guru Nanak Dev Engineering College, Ludhiana

## Department of Electronics and Communication Engineering

B. Tech. (Electronics and Communication Engineering)

**Subject Code: OEEC-111**
**Subject Name: e-Waste Recycling and Management**

| | |
|---|---|
| **Programme:** B.Tech. | **L:** 3 **T:** 0 **P:** 0 |
| **Semester:** 7 | **Teaching Hours:** L: 39 |
| **Theory/Practical:** Theory | **Credits:** 3 |
| **Internal Marks:** 40 | **Percentage of Numerical/Design Problems:** NIL |
| **External Marks:** 60 | **Duration of End Semester Exam (ESE):** 3 hours |
| **Total Marks:** 100 | **Elective Status:** Open Elective Course |

**Prerequisites:** NIL
**Additional Material Allowed in ESE:** NIL

**On Completion of the course, the student will have the ability to:**

| CO# | Course Outcomes | POs | PSOs |
|---|---|---|---|
| 1. | Assess the impact of e-Waste on health and environment | 7(3) | - |
| 2. | Comprehend the legal aspects of e-Waste management | 8(3) | - |
| 3. | Ascertain the responsibility of various stakeholders for e-Waste management | 6(1) | - |
| 4. | Understand the impact of e-Waste mismanagement on environment and health. | 7(2) | - |
| 5. | Explore the distinctive recycling initiatives from around the world | 6(3) | - |
| 6. | Implement recovery techniques related to metals and PCBs e-Waste | 3(2), 4(3) | 1(3),3(2) |

GNDEC

**Detailed Contents:**

**Part A**

**Introduction** **8 hours**

Introduction to e-Waste, Indian and global scenario of e-Waste, e-Waste generation in India, handling and storage of electronic waste until disposal, e-Waste handling solutions for both urban and rural perspectives.

**e-Waste legislation** **8 hours**

Regulatory regime for e-waste in India, Hazardous waste management rules 2003, e-Waste management rules 2015, Regulatory compliance including roles and responsibility of different stakeholders – Producer, Manufacturer, Transporter, Dealer, Consumer, Dismantler, Collection Centre, Recycler, Refurbisher.

**Authorization and certification** **6 hours**

Procedure: grant of authorization, registration with pollution control board, storage of e-Waste, Power to suspend or cancel authorization

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**Part-B**

**WEEE Directives and Recovery Methods** **9 hours**

Introduction, direct and indirect impact on the environment, Hazardous composition in e-Waste, Environmental and Health implications, Reduction in use of hazardous materials in the manufacture of Electrical and Electronic Equipment, Useful material recovery methods: Conventional Vs Green methods, Bioleaching – Process, techniques and applications, Hydrometallurgical techniques for PCB recovery.

**Recycling Technologies** **8 hours**

Modern recycling technologies, Guidelines for environmentally sound management of e-Waste, Environmentally-sound treatment technology for e-Waste, Integrated e-Waste recycling and treatment facility.

**Text Books:**

1. Tchobanoglous G., Theisen H., Viquel S.A., "Integrated Solid Waste Management: Engineering, Principles and Management issues", Tata McGraw Hill Publishing Company Ltd., NewDelhi.
2. Johri R., "e-waste: implications, regulations, and management in India and current global best practices", TERI Press, New Delhi.

**Reference Books:**

1. Peavy H.S., Rowe D.R., Tchobanoglous G., "Environmental Engineering", Tata McGraw Hill Publishing Company Ltd., New Delhi
2. Cunningham W.P., Cunningham M.A., "Principles of Environmental Science", Tata McGraw Hill Publishing Company Ltd., New Delhi.
3. Krishnamoorthy B., "Environmental Management, Text Book and Cases", PHI Learning Ltd., New Delhi.
4. Majeti Narasimha Vara Prasad, Meththika Vithanage, Anwesha Borthakur", Handbook of Electronic Waste Management Ist Edition" Butterworth-Heinemann.

**E books and online learning materials:**

1. https://www.classcentral.com/course/swayam-electronic-waste-management-issues-and-challenges-10111
2. http://www.ppcb.gov.in/Attachments/E%20Waste%20Rules/1st.pdf
3. http://www.ppcb.gov.in/Attachments/E%20Waste%20Rules/guidelines%20E-Waste.pdf
4. http://www.ppcb.gov.in/ewasterules.aspx
5. https://cpcb.nic.in/displaypdf.php?id=RS1XYXN0ZS9FLVdhc3RlTV9SdWxlc18yMDE2LnBkZg==

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07 NOV 2023

**Subject Code: OEME-112**

**Subject Name: Entrepreneurship Practices**

| | |
|---|---|
| **Programme: B.Tech** | **L:3 T:0 P:0** |
| **Semester :6/7/8** | **Teaching Hours: 36hrs** |
| **Theory/Practical: Theory** | **Credits:03** |
| **Internal Marks:40** | **Percentage of Numerical/design/programming problems: 10%** |
| **External Marks:60** | **Duration of End Semester exam(ESE):03hours** |
| **Total Marks:100** | **Elective Status: Open** |

**Prerequisites: Nil**

**Additional Material allowed in ESE: Nil**

On completion of the course, the student will have ability to:

| CO# | Course Outcomes |
|---|---|
| 1 | Comprehend the problems and find a business opportunity |
| 2 | Devise strategies for different types of markets |
| 3 | Present the value proposition and compete in market |
| 4 | Build a Minimum viable product |
| 5 | Understand intricacies of financials in business |
| 6 | Present his pitch deck |

**Part-A**

**Unit 1: Basics of Entrepreneurship** **07 hours**

Problem Identification and Opportunity Discovery: Conduct Opportunity Discovery, Problem Validation, Sharpen the Problem Pitch

Customer and Markets: Identify the Market Type, Explore Market Segment; Determine Market Positioning, Create Customer Persona

**Unit2: Value Proposition & Competition** **07 hours**

Create a Compelling Value Proposition: Craft your core value proposition, Create Sustainable Differentiation Strategy, Deliver Value

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Competitive Advantage: Identify competitors, Identify critical product features, Conduct Feature ranking

**Unit3: Business Models** **07 hours**

Business Model: Build and test a business model, Pivot or Persevere, Identify the riskiest assumptions.

Build your MVP: Build your prototype, Test with early adopters, Conduct Customer Interviews, Refine the Prototype, Build Minimum Viable Product

**Part-B**

**Unit 4:Finance in Business** **07 hours**

Financial Feasibility: Ascertain Costs, Arrive at appropriate pricing strategy, Financial Projections, Key Financial Metrics

Go To market Strategy: Identify the appropriate channels, Build Strategic partnerships, Devise a Market penetration strategy

**Unit 5: Growth & Funding** **08 hours**

Managing growth and Targeting Scale: Growth Plan Structure for the Scaling Strategy Customer acquisition;Enhancing productivity,Process improvements,Operational excellence,Manage money

Funding Strategy: Sources of funding and uses of Funds Statement, Map the Start-up Lifecycle to Funding Options; Valuation Create the Pitch Deck

**Books & Resources:**

All Resources will be provided by Wadhwani Foundation

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**Subject Code: :** OEEE-106
**Subject Name**: *Energy Efficient Machines*

| | |
|---|---|
| **Programme:** B.Tech. | **L:** 3 **T:** 1 **P:** 0 |
| **Semester:** 6 | Teaching **Hours:** 36L |
| **Theory/Practical:** Theory | **Credits:** 3 |
| **Internal Marks:** 40 | **Percentage of Numerical/Design Problems:** 30% |
| **External Marks:** 60 | **Duration of End Semester Exam(ESE):** 3hours |
| **Total Marks:** 100 | **Elective Status:** Open Elective |

**Prerequisites:** Induction Motor Basics

**Additional Material Allowed in ESE:** Scientific Calculator

**On Completion of the course, the student will have the ability to:**

| CO# | Course Outcomes |
|---|---|
| 1. | Understand the importance of energy management and audit |
| 2. | Apply energy conservation measures in industrial and agriculture sector |
| 3. | Understand the design modifications in energy efficient motor as compared to standard motor |
| 4. | Understand the power factor correction of non-linear loads |
| 5. | Evaluate motor efficiency and selection of drive for industrial use |
| 6. | Apply a computational tool to analyze the payback period of energy efficient motors |

**Detailed Contents:**

**PART-A**

**Energy Management and Audit** **(6L)**

Introduction to Energy Management, Energy Audit and its Need, Types of Energy Audit, Energy Conservation in context with induction motor loads in Industrial and Agricultural Sector, Energy Audit Instruments.

**Reactive Power Management in Electrical Systems** **(8L)**

Electricity Billing, Electrical Load Management and Maximum Demand Control, low power factor issues with standard induction motor Methods of Power Factor Improvement, Selection and Location of Capacitors, Performance. Assessment of Power Factor Correction Capacitors, Power Factor with Non-Linear Loads, Harmonics distortion factor and power quality.

**Adjustable Speed Drives** **(4L)**

Introduction to AC drives, AC drive classification. Variable Frequency Drives, drives in industrial automation. AC drive evaluation and selection.



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**PART-B**

**Energy Efficient Motors** **(10L)**

Difference between Standard Motors and Energy Efficient Motors, NEMA Design A, B, C, D, Wound Rotor, Multi speed Motors, Motor Efficiency Determination Methods: Direct Measurement Method, Loss Segregation Method, Motor Efficiency Labeling, Factors for selection of Energy Efficient Motors, Over-Motoring, Eddy Current.

**Economics of Energy Efficient Motors and Systems** **(8L)**

Motor Life Cycle Cost, Direct Saving and Payback Analysis, present Worth Method with constant Power Costs and Increasing over Costs, Introduction to software like MATLAB/ANSYS for various computations related to energy efficient machines and energy audit.

**Text Books:**

1 J. C. Andreas, Energy Efficient Electric Motors, Marcel Dekker Inc.1992

2 T. Albert, Introduction to Efficient Electric System Design, The Fairmount Press Prentice Hall.

3 S.C. Tripathi, Electric Energy Utilization and Conservation, Tata Mc-Graw Hill 1991.

4 B. Charles, Handbook of Modern Electronics and Electrical Engineering, John Wiley & Sons,1986.

**E-Books and online learning material:**

1 . NEMA https://www.nema.org. Accessed on Aug 4 2020

2 . BEE web-link https://www.beeindia.go Accessed on Aug 4 2020

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